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Lion

The **lion** (***Panthera leo***) is a large cat of the genus *Panthera*, native to Africa and India. It has a muscular, broad-chested body; a short, rounded head; round ears; and a dark, hairy tuft at the tip of its tail. It is sexually dimorphic; adult male lions are larger than females and have a prominent mane. It is a social species, forming groups called prides. A lion's pride consists of a few adult males, related females, and cubs. Groups of female lions usually hunt together, preying mostly on large ungulates. The lion is an apex and keystone predator; although some lions scavenge when opportunities occur and have been known to hunt humans, lions typically do not actively seek out and prey on humans.

The lion inhabits grasslands, savannahs, and shrublands. It is usually more diurnal than other wild cats, but when persecuted, it adapts to being active at night and at twilight. During the Neolithic period, the lion ranged throughout Africa and Eurasia, from Southeast Europe to India, but it has been reduced to fragmented populations in sub-Saharan Africa and one population in western India. It has been listed as Vulnerable on the IUCN Red List since 1996 because populations in African countries have declined by about 43% since the early 1990s. Lion populations are untenable outside designated protected areas. Although the cause of the decline is not fully understood, habitat loss and conflicts with humans are the greatest causes for concern.

One of the most widely recognised animal symbols in human culture, the lion has been extensively depicted in sculptures and paintings, on national flags, and in literature and films. Lions have been kept in menageries since the time of the Roman Empire and have been a key species sought for exhibition in zoological gardens across the world since the late 18th century. Cultural depictions of lions were prominent in Ancient Egypt, and depictions have occurred in virtually all ancient and medieval cultures in the lion's historic and current range.

Lion

Temporal range:

Male in Okonjima, Namibia

Female in Okonjima

Conservation status

Vulnerable (IUCN 3.1)[2]

CITES Appendix II (CITES)[a][2]

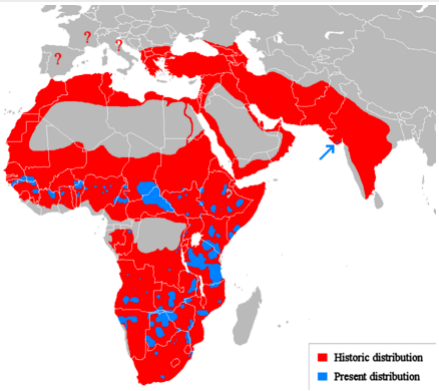
Scientific classification

Domain:	Eukaryota
Kingdom:	Animalia
Phylum:	Chordata
Class:	Mammalia
Order:	Carnivora
Suborder:	Feliformia
Family:	Felidae
Subfamily:	Pantherinae

Etymology

The English word *lion* is derived via Anglo-Norman *liun* from Latin *leōnem* (nominative: *leō*), which in turn was a borrowing from Ancient Greek λέων *léōn*. The Hebrew word לָבִיא *lavi* may also be related.[4] The generic name *Panthera* is traceable to the classical Latin word 'panthēra' and the ancient Greek word πάνθηρ 'panther'.[5]

Genus:	*Panthera*
Species:	***P. leo***[1]
Binomial name	
Panthera leo[1] (Linnaeus, 1758)[3]	
Subspecies	
P. l. leo *P. l. melanochaita* †*P. l. sinhaleyus*	



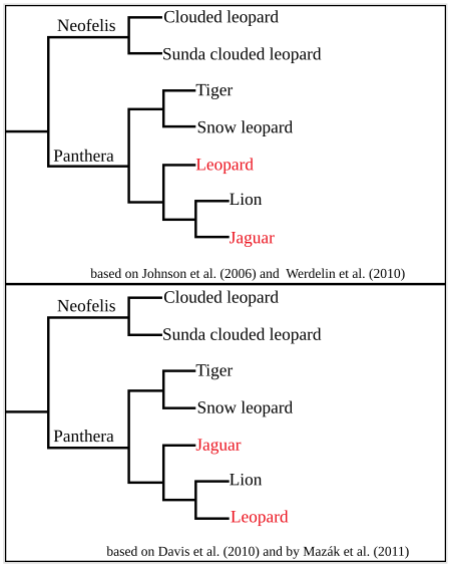
Historical and present distribution of the lion in Africa, Asia and Europe

Taxonomy

Felis leo was the scientific name used by Carl Linnaeus in 1758, who described the lion in his work *Systema Naturae*.[3] The genus name *Panthera* was coined by Lorenz Oken in 1816.[10] Between the mid-18th and mid-20th centuries, 26 lion specimens were described and proposed as subspecies, of which 11 were recognised as valid in 2005.[1] They were distinguished mostly by the size and colour of their manes and skins.[11]

Subspecies

In the 19th and 20th centuries, several lion type specimens were described and proposed as subspecies, with about a dozen recognised as valid taxa until 2017.[1] Between 2008 and 2016, IUCN Red List assessors used only two subspecific names: *P. l. leo* for African lion populations, and *P. l. persica* for the Asiatic lion population.[2][12][13] In 2017, the Cat Classification Task Force of the Cat Specialist Group revised lion taxonomy, and recognises two subspecies based on results of several phylogeographic studies on lion evolution, namely:[14]



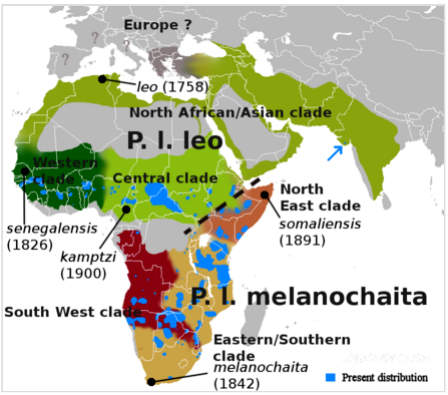
The upper cladogram is based on the 2006 study,[6][7] the lower one on the 2010[8] and 2011[9] studies.

- *P. l. leo* (Linnaeus, 1758) – the nominate lion subspecies includes the Asiatic lion, the regionally extinct Barbary lion, and lion populations in West and northern parts of Central Africa.[14] Synonyms include *P. l. persica* (Meyer, 1826), *P. l. senegalensis* (Meyer, 1826), *P. l. kamptzi* (Matschie, 1900), and *P. l. azandica* (Allen, 1924).[1] Multiple authors referred to it as 'northern lion' and 'northern subspecies'.[15][16]
- *P. l. melanochaita* (Smith, 1842) – includes the extinct Cape lion and lion populations in East and Southern African regions.[14] Synonyms include *P. l. somaliensis* (Noack 1891), *P. l. massaica* (Neumann, 1900), *P. l. sabakiensis* (Lönnberg, 1910), *P. l. bleyenberghi* (Lönnberg, 1914), *P. l. roosevelti* (Heller, 1914), *P. l. nyanzae* (Heller, 1914), *P. l. hollisteri* (Allen, 1924), *P. l. krugeri* (Roberts, 1929), *P. l. vernayi* (Roberts, 1948), and *P. l. webbiensis* (Zukowsky, 1964).[1][11] It has been referred to as 'southern subspecies' and 'southern lion'.[16]

However, there seems to be some degree of overlap between both groups in northern Central Africa. DNA analysis from a more recent study indicates, that Central African lions are derived from both northern and southern lions, as they cluster with *P. leo leo* in mtDNA-based phylogenies

whereas their genomic DNA indicates a closer relationship with P. *leo melanochaita*.[17]

Lion samples from some parts of the Ethiopian Highlands cluster genetically with those from Cameroon and Chad, while lions from other areas of Ethiopia cluster with samples from East Africa. Researchers, therefore, assume Ethiopia is a contact zone between the two subspecies.[18] Genome-wide data of a wild-born historical lion sample from Sudan showed that it clustered with *P. l. leo* in mtDNA-based phylogenies, but with a high affinity to *P. l. melanochaita*. This result suggested that the taxonomic position of lions in Central Africa may require revision.[19]



Range map showing distribution of subspecies and clades

Fossil records

Other lion subspecies or sister species to the modern lion existed in prehistoric times:[20]

Skull of an American lion on display at the National Museum of Natural History

- *P. l. sinhaleyus* was a fossil carnassial excavated in Sri Lanka, which was attributed to a lion. It is thought to have become extinct around 39,000 years ago.[21]
- *P. fossilis* was larger than the modern lion and lived in the Middle Pleistocene. Bone fragments were excavated in caves in the United Kingdom, Germany, Italy and Czech Republic.[22][23]
- *P. spelaea*, or the **cave lion**, lived in Eurasia and Beringia during the Late Pleistocene. It became extinct due to climate warming or human expansion latest by 11,900 years ago.[24] Bone fragments excavated in European, North Asian, Canadian and Alaskan caves indicate that it ranged from Europe across Siberia into western Alaska.[25] It likely derived from *P. fossilis*,[26] and was genetically isolated and highly distinct from the modern lion in Africa and Eurasia.[27][26] It is depicted in Paleolithic cave paintings, ivory carvings, and clay busts.[28]
- *P. atrox*, or the **American lion**, ranged in the Americas from Canada to possibly Patagonia.[29] It arose when a cave lion population in Beringia became isolated south of the Cordilleran Ice Sheet about 370,000 years ago.[30][31] A fossil from Edmonton dates to 11,355 ± 55 years ago.[32]

Evolution

The *Panthera* lineage is estimated to have genetically diverged from the common ancestor of the Felidae around 9.32 to 4.47 million years ago to 11.75 to 0.97 million years ago.[6][33][34] Results of analyses differ in the phylogenetic relationship of the lion; it was thought to form a sister group with the jaguar that diverged 3.46 to 1.22 million years ago,[6] but also with the leopard that diverged 3.1 to 1.95 million years ago[8][9] to 4.32 to 0.02 million years ago. Hybridisation between lion and snow leopard ancestors possibly continued until about 2.1 million years ago.[34] The lion-leopard clade was distributed in the Asian and African Palearctic since at least the early Pliocene.[35] The earliest fossils recognisable as lions were found at Olduvai Gorge in Tanzania and are estimated to be up to 2 million years old.[33]